

Student ID 1 & 2: *(Do this in pairs)*

Edit Distance rules for this worksheet:

- You are given two strings A and B .
- Allowed operations are insert, delete, and replace (each has cost 1).
- The goal is to transform A into B using minimum total cost.

1. Warm-up: compute the distance

For each pair (A, B) , write the minimum edit distance.

A	B	$ED(A, B)$
kitten	sitting	
flaw	lawn	
algorithm	altruistic	
care	score	

2. Choose a DP state

Let $m = |A|$, $n = |B|$, and define a table $ED[0..m][0..n]$. Complete the state definition:

$ED[i][j] =$ _____

3. Fill an Edit Distance table

Use $A = \text{care}$ and $B = \text{score}$. Fill the full DP table $ED[i][j]$.

	ϵ	s	c	o	r	e
ϵ						
c						
a						
r						
e						

4. **Base cases**

What are the base cases $ED[i][0]$ and $ED[0][j]$?

5. **Write the recurrence**

Write the full recurrence for $ED[i][j]$ for both cases: $A[i] = B[j]$ and $A[i] \neq B[j]$.

6. **Complexity**

What is the overall time and space complexity of DP solution based on above recurrence?